Analogue Electronic Devices

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Analogue Electronic Devices (A04587)

Short Title: Analogue Electronic Devices
Faculty Science and Computing
Department Engineering Technology
Cluster Electronics
Author CLAIRE.FITZPATRICK
Credits 5 **Level:** Introductory

Description of Module / Aims

This module is structured to provide students with a comprehensive introduction to analogue electronics. Fundamental semiconductor junction operation and fabrication techniques are introduced along with central analogue devices such as diodes, BJT transistors and FET transistors.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (`WD_EAUTO_B`)	1 / 2 / E
`ELTR-0010`	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	1 / 2 / E
`ELTR-0010`	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	1 / 2 / M
`ELTR-0010`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	1 / 2 / E
`ELTR-0010`	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	1 / 2 / M
`ELTR-0010`	BEng (Hons) (`WD_ETRIC_B`)	1 / 2 / E
`ELTR-0010`	BSc (Hons) in Computer Science (`WD_KCMSC_B`)	1 / 2 / M

Indicative Content

- Fundamental Semiconductor devices theory
- PN junction Operation and Characteristics
- LED's, varactor diodes, zener diodes, photodiodes
- BJT, JFET and MOSFET Transistor operation and characteristics
- Shockly diodes, SCR's. DIAC's, TRIAC's

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the critical properties of semi-conductor material.
2. Summarise the operation and characteristics of PN Junction based devices.
3. Summarise the operation and characteristics of BJT and Field Effect transistors.
4. Analyse fundamental d.c circuit operation for PN junction based devices and transistor type devices.
5. Produce suitable circuit designs for basic switch type circuits.
6. Construct and document the operation of basic diode and transistor circuitry in a laboratory setting.

Learning and Teaching Methods

- Lectures
- Integrated Tutorials
- Laboratory Practicals
- Project Work

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
Lab Report	20%	6
Project	10%	5,6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Bell, D. _Fundamentals of Electronic Devices and Circuits_. UK: Oxford University Press, 2009.
- Bogart, T., J.S. Beasley and G. Rico. _Electronic Devices and Circuits_. USA: Pearson, 2004.
- Malvino, A.P., D. Bates and G. Rico. _Electronic Principles_. 8th ed. .: McGraw Hill, 2015.

Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Electronics